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Roll No

EX-702(C)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

High Voltage Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain how the temperature affects the breakdown strength of solid dielectrics? 7
b) Define wave front time and wave tail time. 7
2. a) Discuss the functions of trigatron gap. 7
b) What is the function of surge arrestor? 7
3. a) List the characteristics of switching surges. 7
b) Define the terms creep age distance an impulse voltage? 7
4. a) State different tests to be conducted on H.V Insulators. 7
b) Briefly explain various numerical methods for estimation of electric field in dielectric materials. 7
5. a) Discuss their relative advantages and disadvantages. 7
b) Explain different insulating materials used in rotating machines. 7

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- 6. a) Define surge voltages. Explain how they are distributed in the windings of power apparatus. 7
- b) Explain the different aspects of insulation design and insulation coordination adopted for EHV systems. 7
- 7. a) What are the mechanisms by which lightning strokes develop an induce over voltages on overhead power lines? Explain. 7
- b) Explain the different electrical tests done on isolators and circuit breakers. 7
- 8. a) Write short notes on Surge current measurement. 7
- b) Classify HV insulating media. 7
